

ANDROMEDA
SURGICAL

LEARNING A BLACK-BOX FUNCTION

Periodic function learning with extrapolation beyond the training range

ML ENGINEERING TAKE-HOME

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RESULT AT A GLANCE

Learn a periodic black-box function on $[-10\pi, 10\pi]$; evaluate at $[50\pi, 60\pi]$ far outside the training range.

Tanh MLP

1.95

Far RMSE

Worse than predicting the mean (0.86)

Snake MLP

0.39

Far RMSE

Keeps the oscillation, but error increases farther from the training window.

Fourier regressor

1.2×10^{-5}

Far RMSE

25 parameters; spectral init + Adam

Message: Matching the model structure to the signal, with a data-driven frequency estimate, is what makes extrapolation work

Separately tuned configurations; seed 42

PROBLEM STATEMENT & SUCCESS CRITERIA

Objective

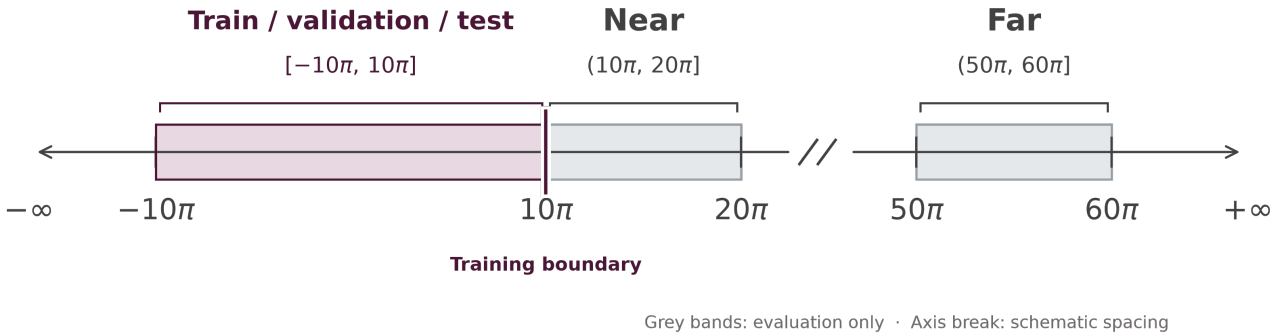
- Learn an approximation $f_\theta(x)$ of a bounded, periodic black-box function from sampled input-output pairs.
- The input domain is unbounded, so the real test is extrapolation beyond the training range.

Constraints

- Training observations restricted to $[-10\pi, 10\pi]$.
- Learn from samples only; no target-function calls during inference.
- No hardcoded target period or frequencies.
- Free: architecture, loss, optimizer, sample count and sampling strategy

Success criteria

- Accurate prediction on held-out in-domain inputs.
- Quantitative evaluation of near and far extrapolation, including failure cases.
- Reproducible training and reliable inference from saved models.
- Clean, modular, tested implementation.



Grey bands: evaluation only · Axis break: schematic spacing

WHY THIS PROBLEM MATTERS IN SURGICAL ROBOTICS

Real-world data constraints

- **Limited data:** expert surgical demonstrations and instrument telemetry are costly to collect.
- **High variability:** anatomy, instruments, surgeon behavior and operating conditions differ between procedures.
- **Safety constraints:** low average error is not enough; failures and uncertainty must be understood.

Connection to the take-home

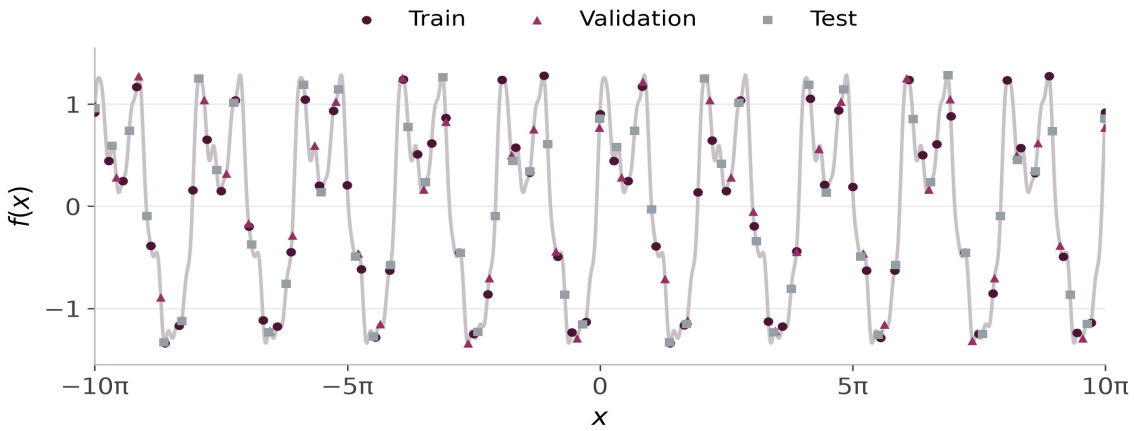
- **Sample efficiency:** how much can we learn from a limited number of observations?
- **Generalization:** what happens when inputs fall outside the training distribution?
- **Model assumption:** can the right mathematical structure make learning more reliable from limited data?

Core question: **How do we build models that generalize reliably when data is scarce and mistakes are costly?**

DATA & EVALUATION DESIGN

In-domain (interpolation)

7,000 train, 1,000 validation, 2,000 test



Subset of the 10,000 samples shown.

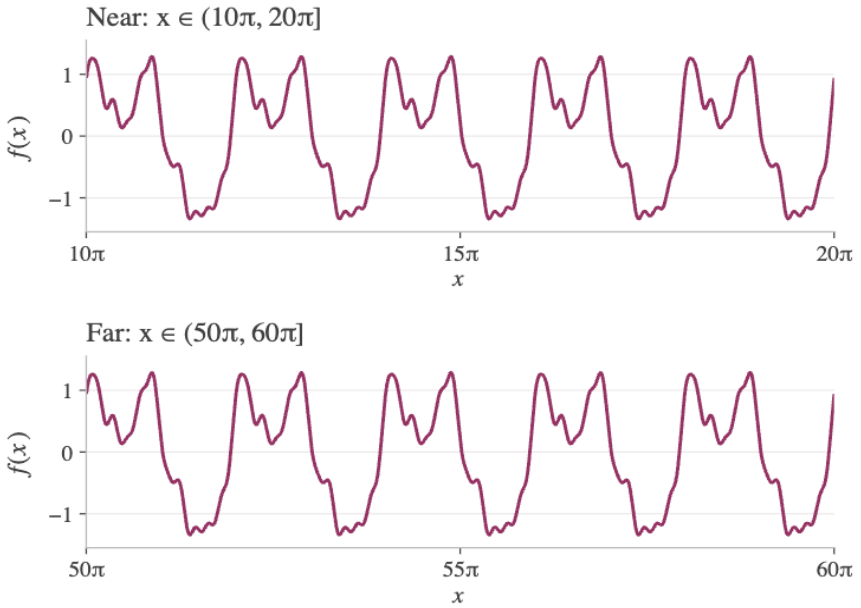
Ground truth function:

$$f(x) = \sin(x) + 0.55\cos(2x) + 0.35\sin(3x + 0.4)$$
$$+0.20\cos(5x - 0.7) + 0.12\sin(7x) + 0.08\cos(11x + 0.2)$$

Period 2π · six frequencies · $\sigma \approx 0.86$, so predicting the mean gives RMSE ≈ 0.86, the reference for all results.

Out-of-domain (extrapolation)

2,000 additional samples per band

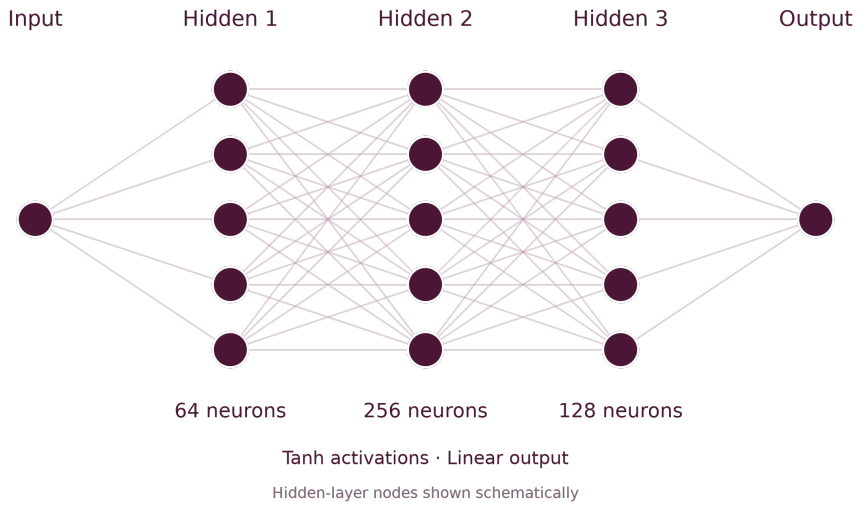


Known to the evaluator only; never for fitting or model selection

Extrapolation samples are used only for evaluation – Not fitting or model selection

MLP REFERENCE CONFIGURATION

Architecture



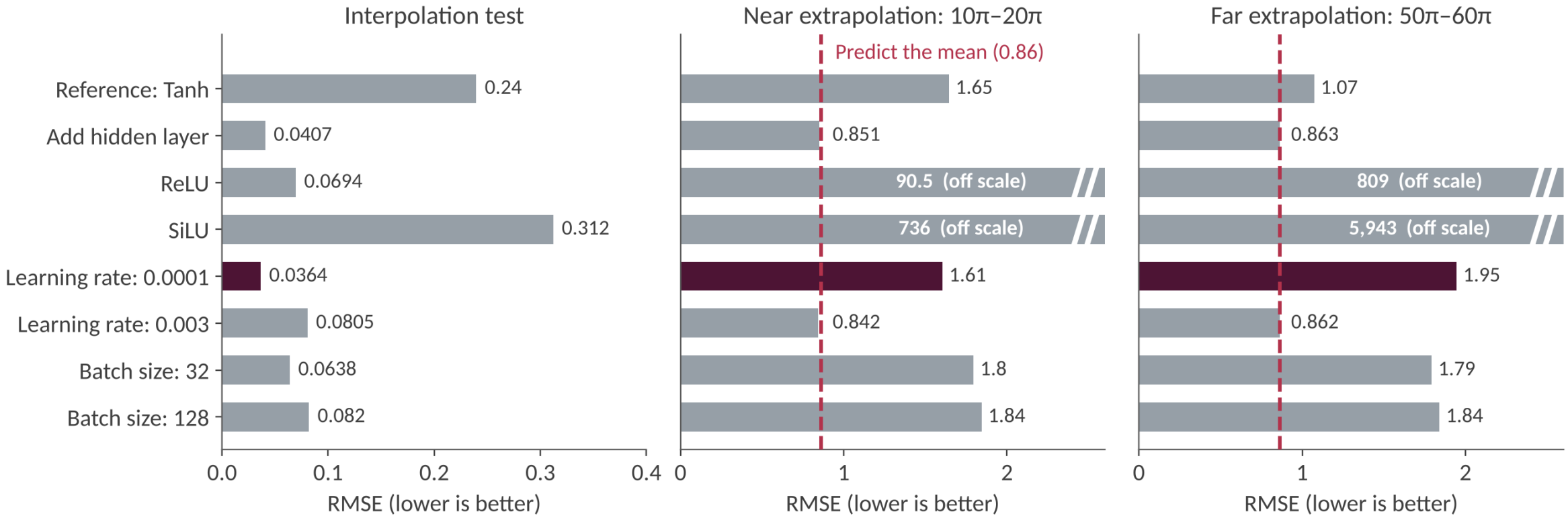
Hypothesis: A MLP can fit the training window but has no mechanism to continue periodic structure beyond it.

After StandardScaler ($\sigma \approx 18.1$) the training window maps to ± 1.7 and the far band to $8.7 - 10.4$, where Tanh saturates and the output flattens.

Training configuration

Parameter	Value
Architecture	$1 \rightarrow 64 \rightarrow 256 \rightarrow 128 \rightarrow 1$
Parameters	49,793
Activation	Tanh
Optimizer	Adam
Learning rate	0.001
Loss	MSE
Early stopping	Validation loss, patience 150
Preprocessing	StandardScaler
Maximum epochs	20,000

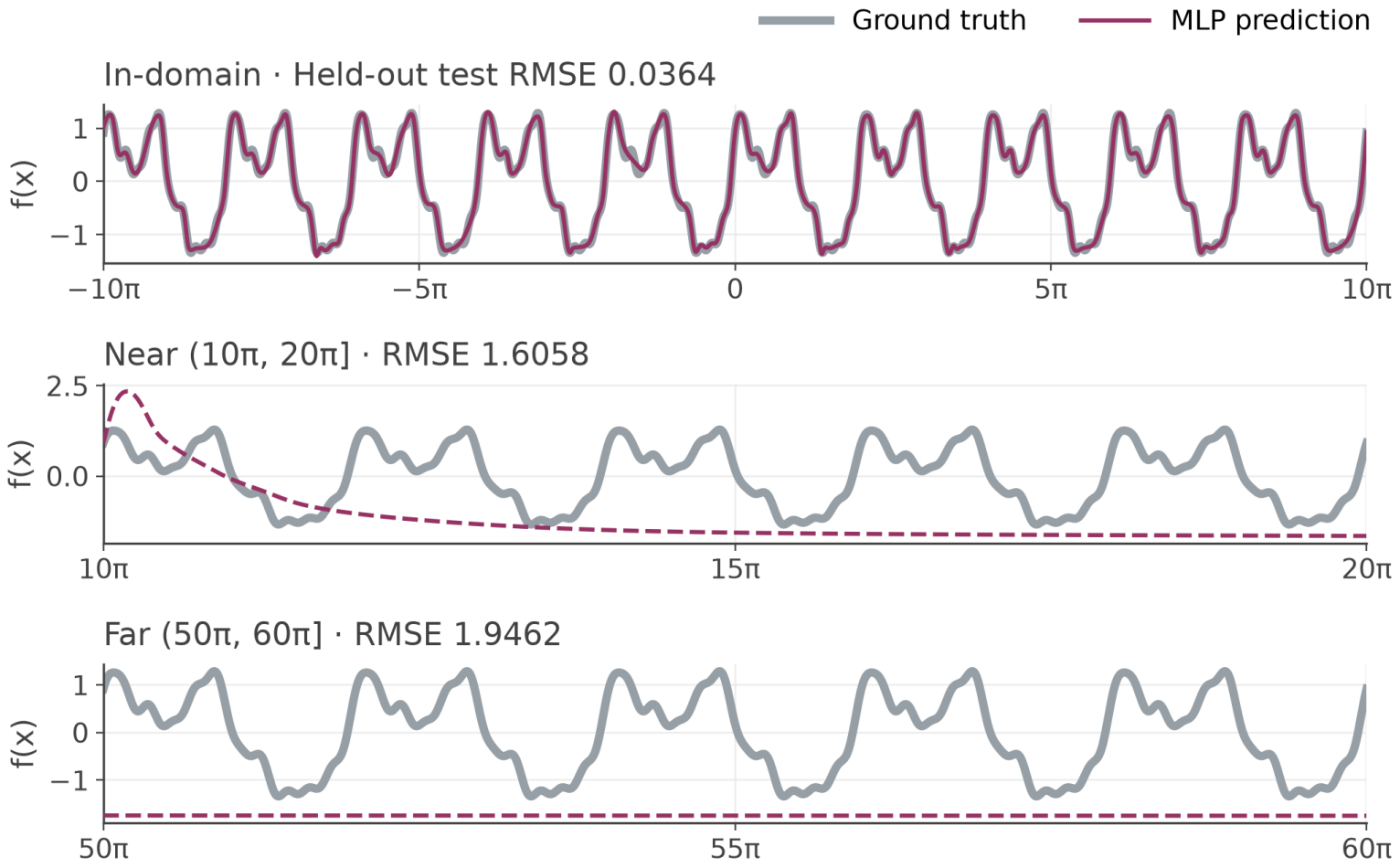
This is the ablation reference; the next slide varies one factor at a time



PERIODIC TARGET: ABLATION RESULTS

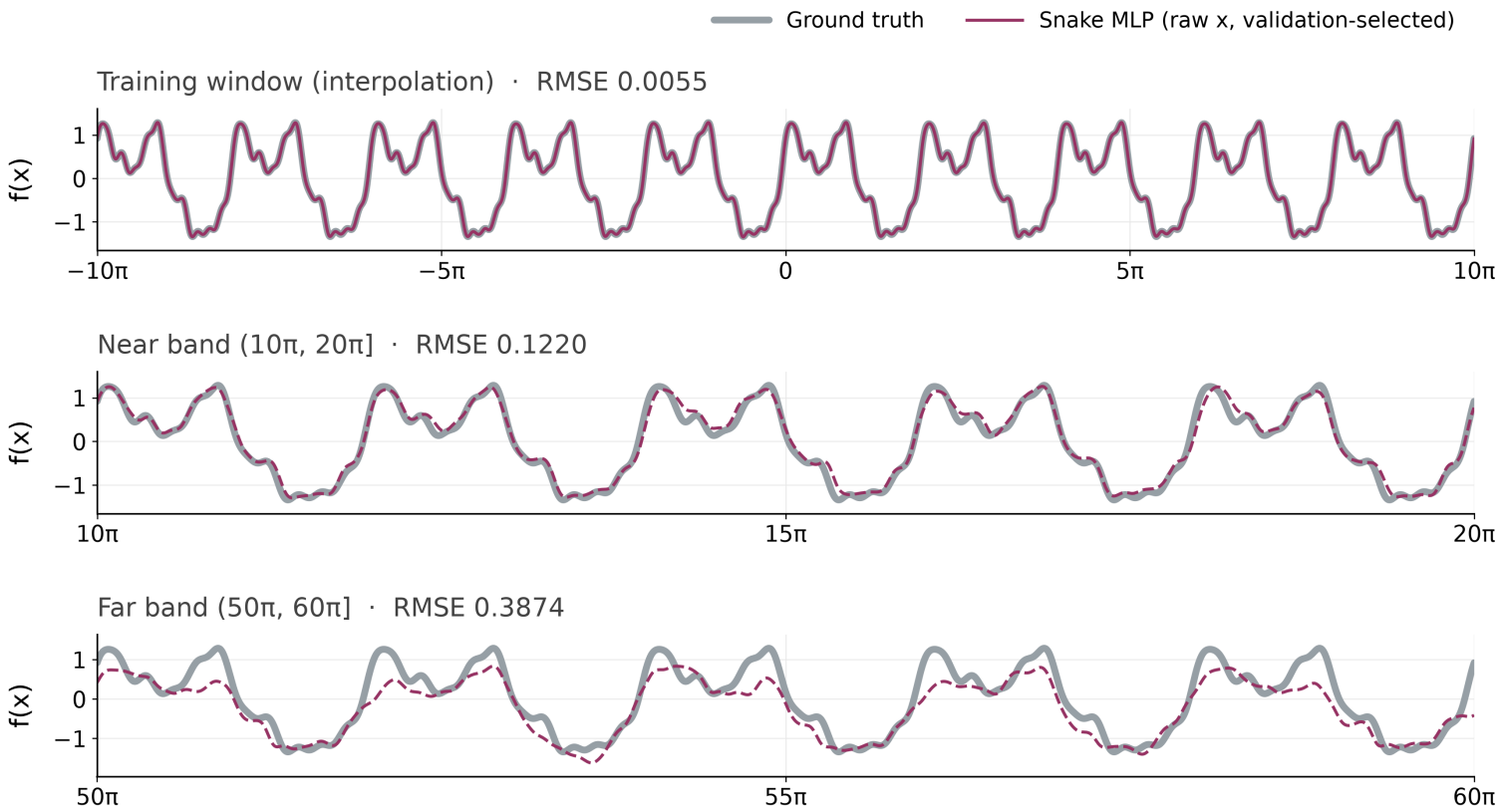
- Selected by validation RMSE: Tanh, learning rate 0.0001
- Eight configurations; one factor changed at a time; seed 42; selection by validation RMSE.
- Interpolation: RMSE 0.0364, R^2 0.9982
- Extrapolation: RMSE 1.61 (near) / 1.95 (far), worse than predicting the mean (0.86)
- ReLU and SiLU diverge off the training range

SELECTED MODEL: INTERPOLATION VS EXTRAPOLATION



- **Selected Model:**
  - Tanh · [64, 256, 128] · Learning rate 0.0001
- **Interpolation:**
  - Closely matches the observed waveform, held out test $RMSE$ 0.0364
- **Near extrapolation:**
  - Deviates beyond the training boundary and approaches a constant; near RMSE 1.6058.
- **Far extrapolation:**
  - Flat output at about -1.75, no oscillation; $RMSE$ 1.95, worse than predicting the mean (0.86)
- **Takeaway:**
  - **Accurate interpolation does not establish reliable periodic extrapolation.**

SNAKE ACTIVATION MLP



- **Model:**
  - Snake MLP · [64, 256, 128] · raw X
- **Interpolation:**
  - Closely matches the waverform; held-out test RMSE 0.0055
- **Near extrapolation:**
  - Follows the waveform past the boundary; RMSE 0.122
- **Far extrapolation:**
  - Oscillations persist, but peak amplitudes and waveform details deviate; RMSE 0.387 (Tanh 1.95; mean 0.86)
- **Takeaway:**
  - The selected Snake MLP retained oscillatory structure beyond training, but error increased across the evaluated bands.

Snake: $x + sin^2(a \cdot x)/a$, Trainable a per neuron · Adam LR 0.001 for weights and a · Batch 128 · Maximum 8,000 epochs · Patience $1{,}500$ · Seed 42

WHY A FOURIER REGRESSOR?

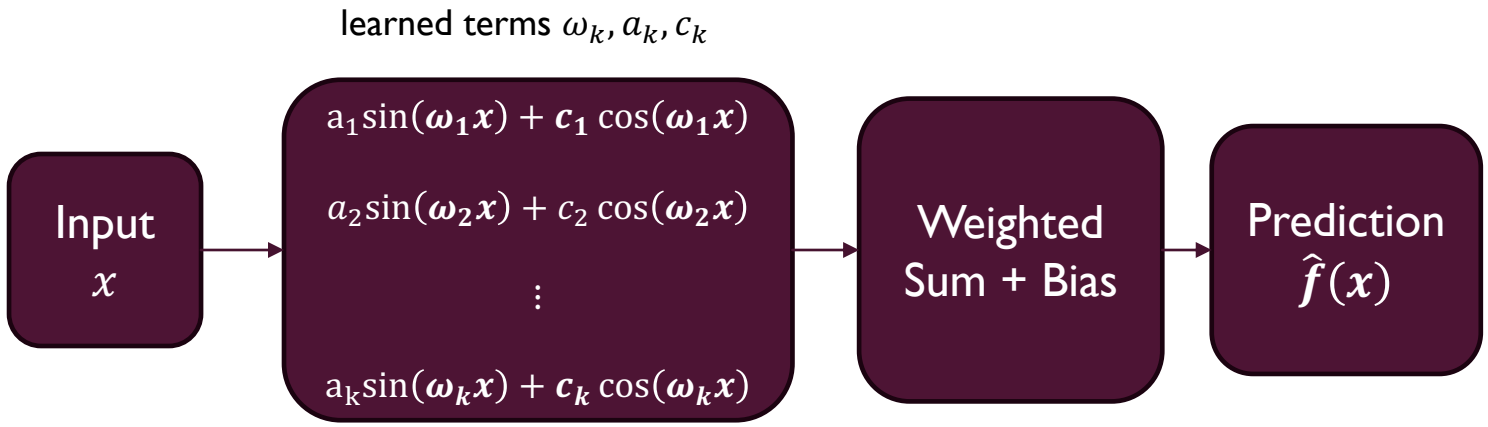
Motivation

- The selected MLP interpolated accurately but flattened outside the training window.
- Snake continued the oscillations, but error still grew with distance.
- A sum of sinusoids builds persistent oscillatory components into the representation.

Model

$$\hat{f}(x) = b + \sum_{k=1}^{K} [a_k \sin(\omega_k x) + c_k \cos(\omega_k x)]$$

- Learn frequencies, sin/cos weights, and bias from samples.
- No target period or frequency is hardcoded.
- K = 8: 8 frequencies + 16 amplitudes + 1 bias = 25 trainable parameters.



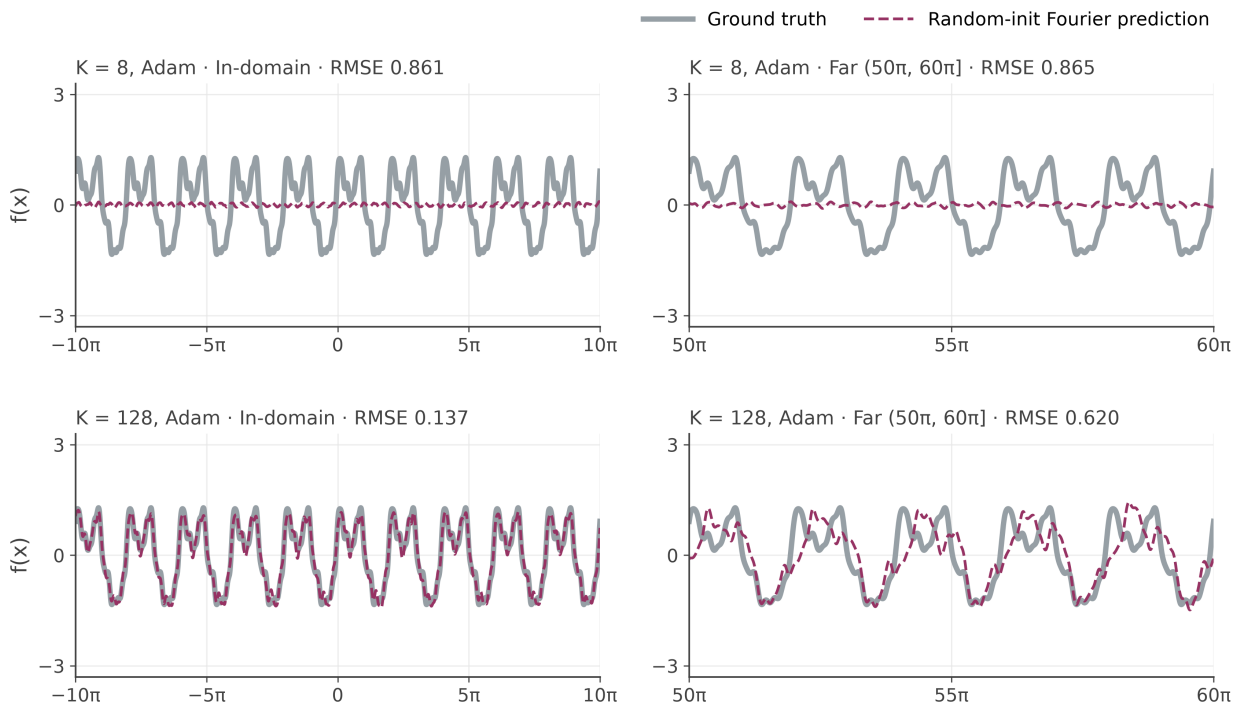
Why it can extrapolate

A frequency error $\delta\omega$ becomes a phase error $\delta\omega \cdot x$, so accuracy far from the training data depends on how precisely the frequencies are recovered.

Frequencies are searched within $\omega \in [0.1, 20]$ (a stated assumption)

Note: 20 is an assumed search bandwidth.

RANDOM INITIALIZATION – EXPERIMENTS & FINDINGS



- **Question**
  - Can gradient-based optimization learn the frequencies from random starting values?
- **Tested**
  - Adam(lr $0.001, 0.003, 0.01$); uniform vs log-uniform random frequencies; $K = 8\ to\ 128$

Configuration	Params	Val	Near	Far
Uniform $K = 8$, Adam	25	0.8610	0.8654	0.8654
Log-uniform $K = 128$, Adam	385	0.1375	0.2401	0.6202
Predict the mean	–	0.86	0.86	0.86

Takeaway: The tested random-initialization configurations did not achieve the spectral model's accuracy: the same-size model ($K = 8$) collapses to the mean, and a $15\times$ larger one ($K = 128$) fit the window but its phase error grows with distance.

SPECTRAL INITIALIZATION – ESTIMATING FREQUENCIES

Motivation

- Random initialization did not recover the frequencies (previous slide): K= 8 collapsed to the mean, and K=128 had phase error that grew with distance.
- Instead, Fit a sinusoid to the unexplained signal:

$$\boldsymbol{E(\omega) = \min_{a,c,b} \sum_{i=1}^{N} [r_i - b - a\sin(\omega x_i) - c\cos(\omega x_i)]^2}$$

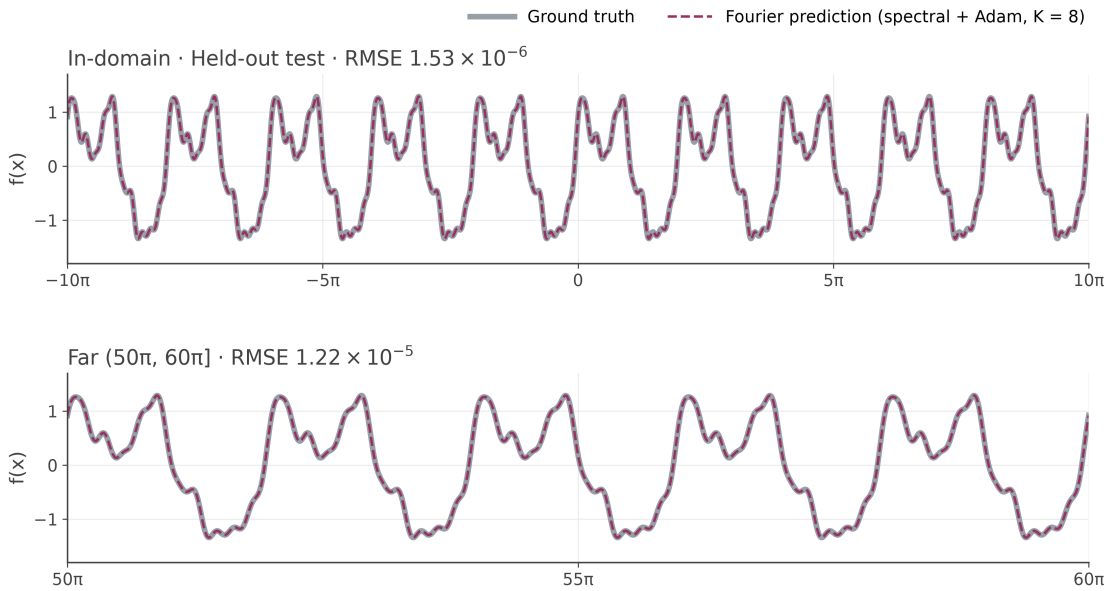
- Here, r_i is the training residual over N samples; initially, $r_i = y_i - \bar{y}_i$ (the mean of y)

Method

- **Score:** Use Lomb–Scargle on the residual, 4,096-point grid over $[0.1, 20]$.
- **Select:** Strongest peak not within the minimum separation ($\approx 2\pi/span$) of a chosen frequency.
- **Refine:** Bounded local search around that grid point.
- **Refit:** All sine/cosine coefficients and bias against the original training labels.
- **Repeat:** Update the residual; K=8 rounds.
- **Fine-tune:** Adam then updates frequencies, coefficients, and bias together.

Takeaway: Starting frequencies are learned from training samples, no target period or frequencies are supplied.

FOURIER REGRESSOR: RESULTS



Configuration	Params	Val	Interp.	Near	Far
Spectral only K=8	25	0.01073	0.01073	0.04804	0.14981
Spectral + Adam K=4	13	0.10193	0.10198	0.10244	0.10902
Spectral+Adam,K=8	**25**	$\mathbf{1.50\times10^{-6}}$	$\mathbf{1.53\times10^{-6}}$	$\mathbf{3.37\times10^{-6}}$	$\mathbf{1.22\times10^{-5}}$
Spectral + Adam K=16	49	4.83×10^{-6}	4.87×10^{-6}	4.54×10^{-5}	0.00154

- Adam substaintially, improved the K=8 spectral fit by about four order of magnitude (far $0.15 \rightarrow 1.2\times10^{-5}$).
- K = 4 underfit this target, increasing from K=8 to K=16 did not improve the measured results.
- K=8 (25 parameters) selected using validation RMSE only.

Takeaway: Spectral initialization plus Adam accurately reproduced the waveform inside training and across both evaluated extrapolated bands.

Note: Seed 42 · Adam learning rate 0.001 · Maximum 30,000 epochs · Patience 3,000 · One noise-free target

FITTED MODEL: 25 PARAMETERS AND THE EFFECT OF ADAM

- Model (K = 8)

$$\hat{f}(x) = b + \sum_{k=1}^{K} [a_k \sin(\omega_k x) + c_k \cos(\omega_k x)]$$

- 25 parameters = 8 frequencies + 8 sine weights + 8 cosine weights + 1 bias.

Fitted function

$$\begin{aligned}\hat{f}(x) = &-1.3422954\times10^{-7} \\ &+0.99442613\sin(0.999999225x) + 0.000201204\cos(0.999999225x) \\ &+0.002763405\sin(1.000118971x) - 0.000102316\cos(1.000118971x) \\ &+0.002809668\sin(1.000135541x) - 0.000098767\cos(1.000135541x) \\ &+0.000000247\sin(2x) + 0.550000012\cos(2x) \\ &+0.322371513\sin(3x) + 0.136296481\cos(3x) \\ &+0.128843769\sin(5x) + 0.152968541\cos(5x) \\ &+0.119999841\sin(7x) + 0.000000163\cos(7x) \\ &-0.015893383\sin(11x) + 0.078405365\cos(11x)\,.\end{aligned}$$

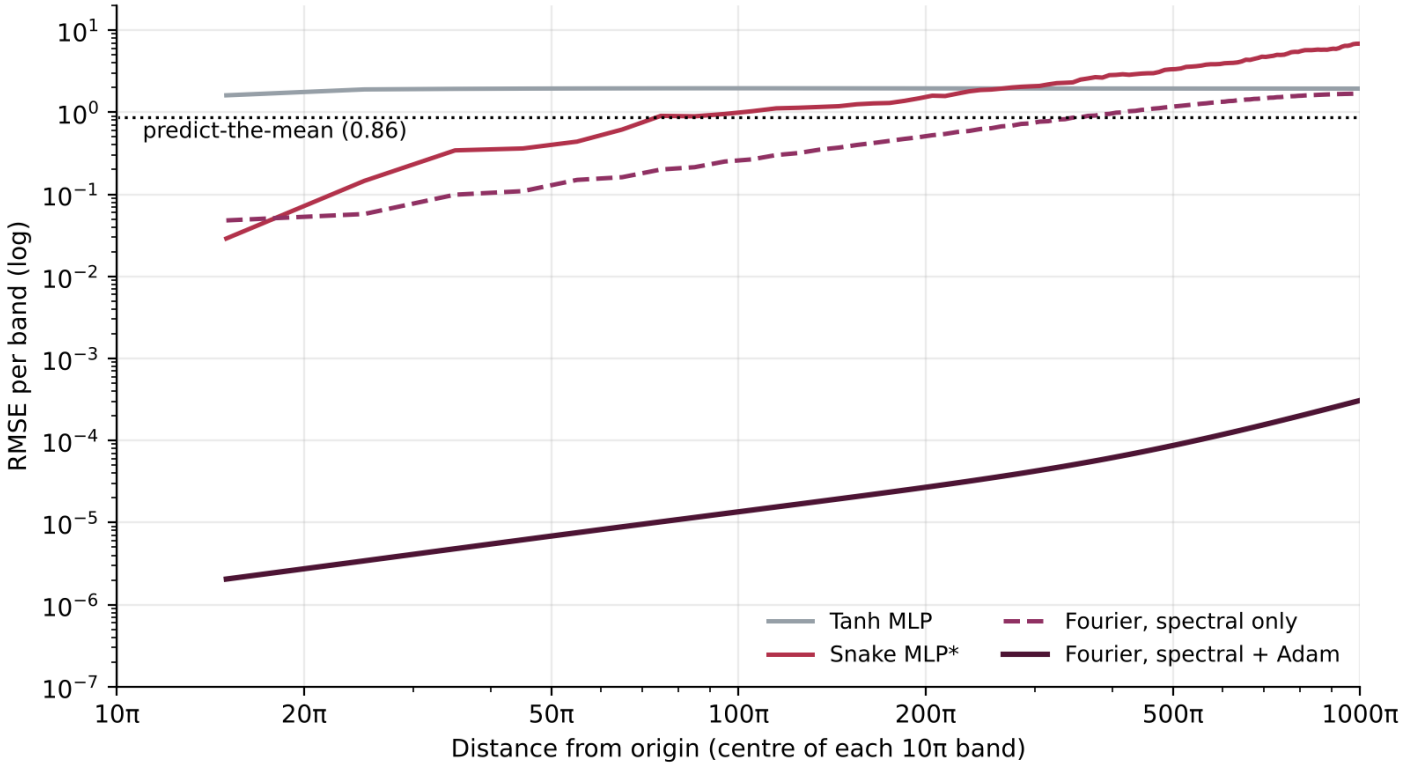
Fitted frequencies ω_k

True ω	Spectral only	Spectral +Adam
1	0.99900	0.999999
1 $(spare)$	0.90183	1.000119
1 $(spare)$	1.09621	1.000136
2	1.99922	2.000000
3	2.99884	3.000000
5	4.99926	5.000000
7	7.00015	7.000000
11	10.99999	11.000000

- Adam closely aligns the six principal frequencies with the target frequencies; two small additional components remain near $\omega = 1$.

Takeaways: Adam refines the frequency by four orders of magnitude, and that is what makes the far band accurate (RMSE $0.150 \rightarrow 1.2\times10^{-5}$).

ERROR VERSUS DISTANCE



Note: the log axis; each gridline is a factor of ten.

Reading the plot

- Tanh: flat near 1.9, worse than predicting the mean (0.86)
- Snake: best MLP in-window, but error grows and passes the mean line near 75π.
- Fourier, spectral only: limited by $\sim 10^{-3}$ frequency error; passes the mean line near 350π.
- Fourier, spectral + Adam: $2\times10^{-6} \rightarrow 3\times10^{-4}$ over 1000π; still a clean $\delta\omega \cdot x$ growth.

Takeaway: At 1000π the Fourier model is $\sim 10^{4}\times$ better than the mean predictor; its error still grows with distance.

ENGINEERING: TESTS, REPRODUCIBILITY AND COMMANDS

What is in the repository

- 229 automated tests across 17 files: data generation, model, spectral initialization, checkpoints, inference, training safety, tracking.
- Composed YAML configs (base, data, model, experiment, ablation): every run is described by files, not code edits.
- Locked environment: pyproject.toml and uv.lock, Python ≥ 3.11.
- Seeded runs; fixed data grids; each run saves checkpoint, preprocessor, prediction and metrics.
- Inference loads a checkpoint only: no access to the target function.
- Mlflow tracking and ruff linting.

Commands

```
# setup and tests
uv sync –locked –inexact
uv run pytest

# train one experiment
uv run python –m scripts.run_experiments \
     --config
configs/experiments/fourier_rich_periodic.yaml

# ablations and multi-seed
uv run python –m scripts.run_fourier_ablation
uv run python –m scripts.run_fourier_multiseed

 # error versus distance
uv run python –m
scripts.evaluate_extrapolation_distance –max-pi 1000
```

Takeaways: Runs are config-driven, seeded and tested, so results regenerate from a locked environment with a few commands

LIMITATION & NEXT STEPS

Limitations

- One noise-free target; real sensor data would add noise and drift.
- Search bound $\omega \leq 20$ is an assumptions; a component above it would be missed.
- Error still grows with distance: a frequency error $\delta\omega$ gives a phase error $\delta\omega \cdot x$ ($far\ RMSE\ 1.2\times 10^{-5}\ at\ 60\pi$)
- Snake and random-init runs are exploratory: one seed and unequal budgets; snake is sensitive to the training trajectory.

Next Steps

- Add noise and evaluate over more seeds for every model, with matched training budgets.
- Derive the frequency bounds from the sampling interval (Nyquist) instead of fixing 20.
- Select K with a held-out criterion rather than a fixed grid of 4, 8, 16.



APPENDIX

SNAKE MLP CONFIGURATION

- Snake activation: $x + \sin^2(a * x)/a$
- a is learned per neuron, initialized at 1.0
- Identity term keeps gradients healthy; the sine term adds periodic structure
- Architecture: $1 \rightarrow 64 \rightarrow 256 \rightarrow 128 \rightarrow 1$
- Parameters: $50{,}241$ ($49{,}793$ weights and biases, plus 448 learnable a values

Parameter	Value
Architecture	$1 \rightarrow 64 \rightarrow 256 \rightarrow 128 \rightarrow 1$
Parameters	$50{,}241$
Activation	Snake (learnable a, $a_0 = 1.0$)
Optimizer	Adam
Learning rate	0.001 (weights and a)
Loss	MSE
Early stopping	Validation loss, patience 1500
Preprocessing	None (raw x)
Maximum epochs	8,000
Batch size	128
Seed	42

Differences from the Tanh reference: *raw x instead of StandardScaler, batch 128 instead of 64, patience 1500 instead of 150. Snake's validation curve is noisy and plateau-prone, so patience 150 stops too early.*

Ziyin, Hartwig, Ueda, "Neural Networks Fail to Learn Periodic Functions and How to Fix It", NeurIPS 2020. paper